

Fixed Income + Derivatives Quant Risk

Step-by-Step Summary

SECTION 1 — CONFIGURATION

1. Model Selection & Parameters

Choose Vasicek or Hull-White; set parameters: κ (mean-reversion speed), θ (long-run mean), σ (volatility), r_0 (initial rate).

2. Simulation Settings

10,000 paths | 252 time steps | 1-day holding period | 99% confidence level.

3. Yield Curve

Fetch live data via FRED API; if no API key is available, fall back to a built-in curve spanning 0.25 yr to 30 yr.

4. Stress Scenarios

Six scenarios: parallel shift, curve twist, credit spread widening, and recession flight-to-quality.

5. Backtesting Configuration

250-day rolling window; significance level $\alpha = 1\%$.

SECTION 2 — NELSON-SIEGEL YIELD CURVE

1. Model Formula

The yield curve fitting model uses four parameters:

- β_0 — long-run level
- β_1 — slope
- β_2 — curvature
- λ — decay factor

Full formula:

$$y(\tau) = \beta_0 + \beta_1 * [(1 - \exp(-\lambda\tau)) / (\lambda\tau)] + \beta_2 * \{[(1 - \exp(-\lambda\tau)) / (\lambda\tau)] - \exp(-\lambda\tau)\}$$

2. Parameter Estimation

The Nelder-Mead optimization algorithm minimizes the sum of squared errors between observed market tenors and model-fitted yields, producing four optimal parameters.

3. Output Interpolation Function

Returns a function that smoothly interpolates and reasonably extrapolates to any maturity τ , avoiding end-point distortion.

SECTION 3 — RATE MODELS

1. Vasicek — Exact Discretization

Uses the exact conditional Gaussian solution, eliminating Euler discretization error. Single-step standard deviation correction:

```
sigma_dt = sqrt[ sigma^2 / (2*kappa) * (1 - exp(-2*kappa*dt)) ]
```

2. Hull-White Model

$\theta(t)$ is dynamically calibrated so the model exactly fits the current market forward rate curve, ensuring no-arbitrage pricing.

3. Antithetic Variates

Paired random draws Z and $-Z$ reduce variance by approximately 50% at virtually zero additional computational cost.

4. `simulate_horizon()`

Directly samples the rate distribution at the end of the holding period, bypassing the intermediate 252-step path simulation — approximately 250× faster.

5. `discount_factor_batch()`

Vectorized batch computation of zero-coupon bond (ZCB) prices across multiple tenors, significantly improving pricing efficiency.

SECTION 4 — DATA PIPELINE

1. Portfolio Data (Simulated SQL Table)

Contains 12 instruments: UST, Corp IG, Corp HY, MBS, ABS Auto, ABS Card, CLO AAA, CLO BBB, etc.

Each position records: notional, coupon, maturity, credit spread, DV01, effective duration, and other fields.

2. Historical P&L Generation

Synthesizes a 250-day historical P&L series: a mixture of a normal distribution ($\sigma \approx \$185K$) and a t-distribution (df = 5, fat tails), simulating the leptokurtic, heavy-tailed characteristics of real-world P&L — used for backtesting.

SECTION 5 — RISK ENGINE

1. Full Repricing

Each Monte Carlo path triggers a full portfolio reprice. Plain bonds use a closed-form ZCB formula; structured products (e.g., ABS) use an effective-duration linear approximation.

2. PSA Prepayment Model (MBS)

Rate-sensitive prepayment model:

$$\text{CPR}_{\text{eff}} = \text{CPR}_{\text{base}} * (1 + \text{beta_refi} * (r_0 - r)), \text{beta_refi} = 8.0$$

Includes a PSA ramp-up period (linear growth over months 1–30); monthly cash flows are generated and discounted to produce the MBS price.

3. VaR / CVaR Calculation

VaR = 1st percentile of the P&L distribution.

CVaR = conditional expectation of losses beyond VaR (always $\geq$ VaR).

4. Stress Testing

Each of the six scenarios refits the Nelson-Siegel curve and performs a full portfolio reprice; output is the P&L shock for each scenario.

5. Key Rate Duration (KRD)

An isolated +1 bp shock is applied at each key tenor to compute KR01 and KRD, capturing twist and butterfly curve risks.

SECTION 6 — MODEL VALIDATION

1. Kupiec Proportion of Failures (POF) Test

H_0 : actual exception rate = expected exception rate (1%). The LR statistic follows $\chi^2(1)$; reject the model if $p < 0.05$.

2. Christoffersen Conditional Coverage (CC) Test

Jointly tests coverage rate and independence of exception clustering. Builds a transition matrix; LR_{CC} follows $\chi^2(2)$.

3. Basel Traffic Light Approach

Over a 250-day window: 0–4 exceptions = Green Zone; 5–9 = Yellow Zone; 10+ = Red Zone. Directly affects the regulatory capital multiplier.

4. Historical Simulation Benchmark

Compares HS VaR/CVaR against Monte Carlo results to quantify model deviation.

5. Sensitivity Analysis

Key parameters σ and κ are perturbed ($\times 0.75/\times 1.25$ and $\times 0.50/\times 1.50$); outputs are $\Delta\text{VaR}\%$ and $\Delta\text{CVaR}\%$, verifying model robustness.

SECTION 7 — CHARTS

- P&L Distribution — histogram with normal fit overlay and VaR/CVaR vertical lines
- Rate Paths — 200 sample paths + mean path + 5%–95% confidence band
- Stress Test Bar Chart — P&L shocks for 6 scenarios (red/green color coding)
- Sensitivity ΔVaR Bar Chart
- KRD Profile — dual-panel bar chart of KRD and KR01 by tenor
- $\alpha \times \sigma$ VaR Heatmap — 7 volatility levels $\times$ 5 confidence levels

SECTION 8 — SR 11-7 DOCUMENTATION

1. Auto-Generated Compliance Document

All calculation results, parameters, and validation outputs are populated into an SR 11-7 standard template and saved as *model_risk_documentation.md*.

2. Document Structure

- Executive Summary
- Model Description & Parameters
- Assumptions & Limitations
- Three-Pillar Validation Results (Backtesting / Benchmark / Sensitivity)
- Governance Framework (Three Lines of Defense, Tier 1 Classification, Annual Review)
- Document Control

3. Attestation

This model has not yet undergone formal independent MRM validation. Production deployment requires CRO approval and registration in the Model Inventory.
